Annex III. Debt Sustainability Analysis

Korea Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

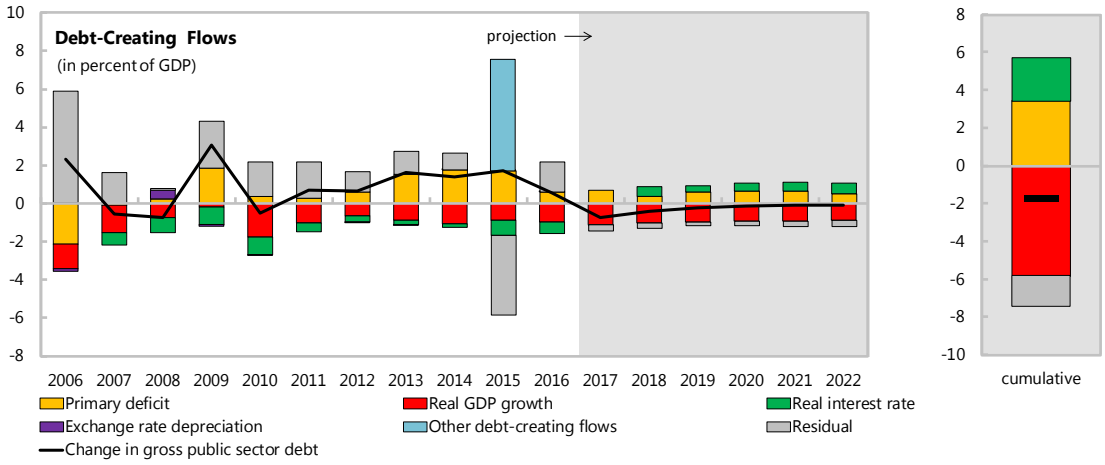
	Actual			Projections					
	2006-2014 [2/]	2015	2016	2017	2018	2019	2020	2021	2022
Nominal gross central government debt	30.0	35.6	36.1	35.4	35.0	34.8	34.6	34.5	34.4
Gross financing needs	5.5	6.8	5.7	1.3	1.5	2.2	2.2	2.7	2.7
Real GDP growth (in percent)	3.7	2.8	2.8	3.2	3.0	2.9	2.8	2.7	2.7
Inflation (GDP deflator, in percent)	1.8	2.4	1.8	2.4	1.5	2.0	2.2	2.1	2.1
Nominal GDP growth (in percent)	5.5	5.3	4.7	5.8	4.6	4.9	5.0	4.9	4.8
Effective interest rate (in percent) [4/]	4.0	2.8	2.5	2.5	3.1	3.1	3.4	3.6	3.9

As of September 27, 2017

Sovereign Spreads		
EMBIG (bp) 3/		238
5Y CDS (bp)		75
Ratings	Foreign	Local
Moody's	Aa2	Aa2
S&Ps	AA	AA
Fitch	AA-	AA-

Contribution to Changes in Public Debt

	Actual			Projections							
	2006-2014	2015	2016	2017	2018	2019	2020	2021	2022	cumulative	debt-stabilizing primary balance [9/]
Change in gross central government debt	0.9	1.7	0.6	-0.7	-0.4	-0.2	-0.1	-0.1	-0.1	-1.7	
Identified debt-creating flows	-1.0	5.9	-1.0	-0.4	-0.1	0.0	0.1	0.2	0.2	-0.1	
Primary deficit	0.5	1.7	0.6	0.7	0.4	0.6	0.7	0.6	0.5	3.4	-0.3
Primary (noninterest) revenue and grants	17.1	16.3	17.2	17.3	17.3	17.3	17.2	17.1	17.1	103.4	
Primary (noninterest) expenditure	17.6	18.0	17.8	17.9	17.7	17.9	17.9	17.8	17.7	106.9	
Automatic debt dynamics [5/]	-1.5	-1.7	-1.6	-1.1	-0.5	-0.6	-0.5	-0.4	-0.3	-3.5	
Interest rate/growth differential [6/]	-1.5	-1.7	-1.6	-1.1	-0.5	-0.6	-0.5	-0.4	-0.3	-3.5	
Of which: real interest rate	-0.5	-0.8	-0.6	0.0	0.5	0.4	0.4	0.5	0.6	2.3	
Of which: real GDP growth	-1.0	-0.9	-1.0	-1.1	-1.0	-1.0	-0.9	-0.9	-0.9	-5.8	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	5.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Net privatization proceeds (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	5.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	1.9	-4.2	1.6	-0.3	-0.3	-0.2	-0.2	-0.3	-0.3	-1.6	



Source: IMF staff.

1/ Public sector is defined as central government.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

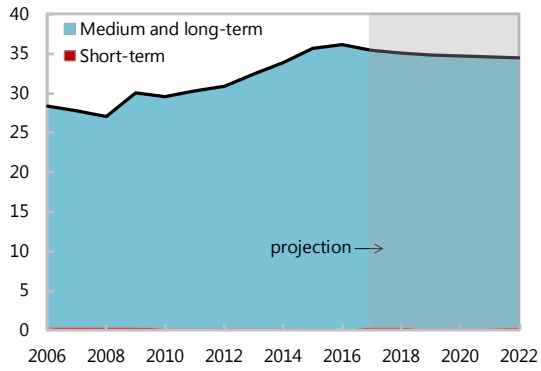
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Korea Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

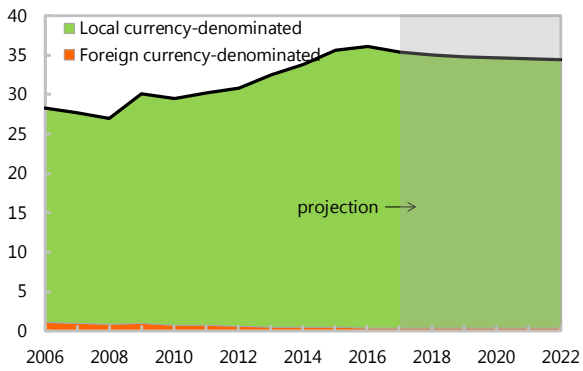
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

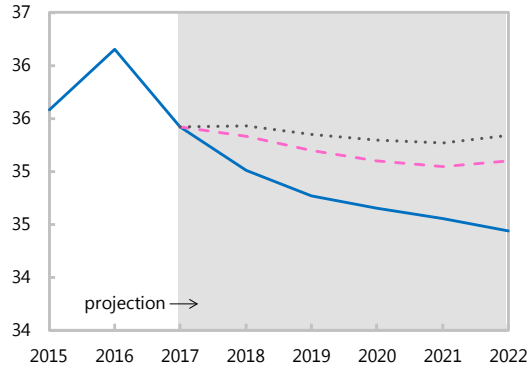


Alternative Scenarios

Baseline — Historical — Constant Primary Balance

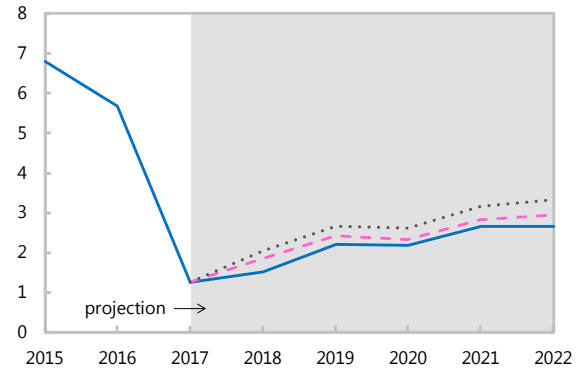
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2017	2018	2019	2020	2021	2022
Real GDP growth	3.2	3.0	2.9	2.8	2.7	2.7
Inflation	2.4	1.5	2.0	2.2	2.1	2.1
Primary Balance	-0.7	-0.4	-0.6	-0.7	-0.6	-0.5
Effective interest rate	2.5	3.1	3.1	3.4	3.6	3.9
Constant Primary Balance Scenario						
Real GDP growth	3.2	3.0	2.9	2.8	2.7	2.7
Inflation	2.4	1.5	2.0	2.2	2.1	2.1
Primary Balance	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Effective interest rate	2.5	3.1	3.1	3.4	3.6	3.9

Historical Scenario	2017	2018	2019	2020	2021	2022
Real GDP growth	3.2	3.3	3.3	3.3	3.3	3.3
Inflation	2.4	1.5	2.0	2.2	2.1	2.1
Primary Balance	-0.7	-0.9	-0.9	-0.9	-0.9	-0.9
Effective interest rate	2.5	3.1	3.1	3.4	3.7	4.0

Source: IMF staff.

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